Airbnb NYC Pricing & Market Analysis

1. Project Overview

This project analyzes the New York City Airbnb Open Data (2019) dataset to identify factors associated with listing prices and compare pricing patterns across room types and locations.

Dataset: NYC Airbnb Open Data, 2019 — **48,895 listings and 16 variables.** The dataset includes price, room type, neighbourhood, minimum nights, reviews, and availability.

Research question: What factors are associated with Airbnb listing prices in New York City?

Tools: Python (Pandas, Matplotlib), MySQL/SQL, Power BI, and statistical regression.

Dataset source: https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data

2. Data Preparation

The dataset was inspected in Python to review its structure, data types, missing values, and duplicates. There were no completely duplicated rows. The columns last_review and reviews_per_month each contained 10,052 missing values, corresponding to listings with no reviews. These records were retained because the absence of reviews is meaningful information.

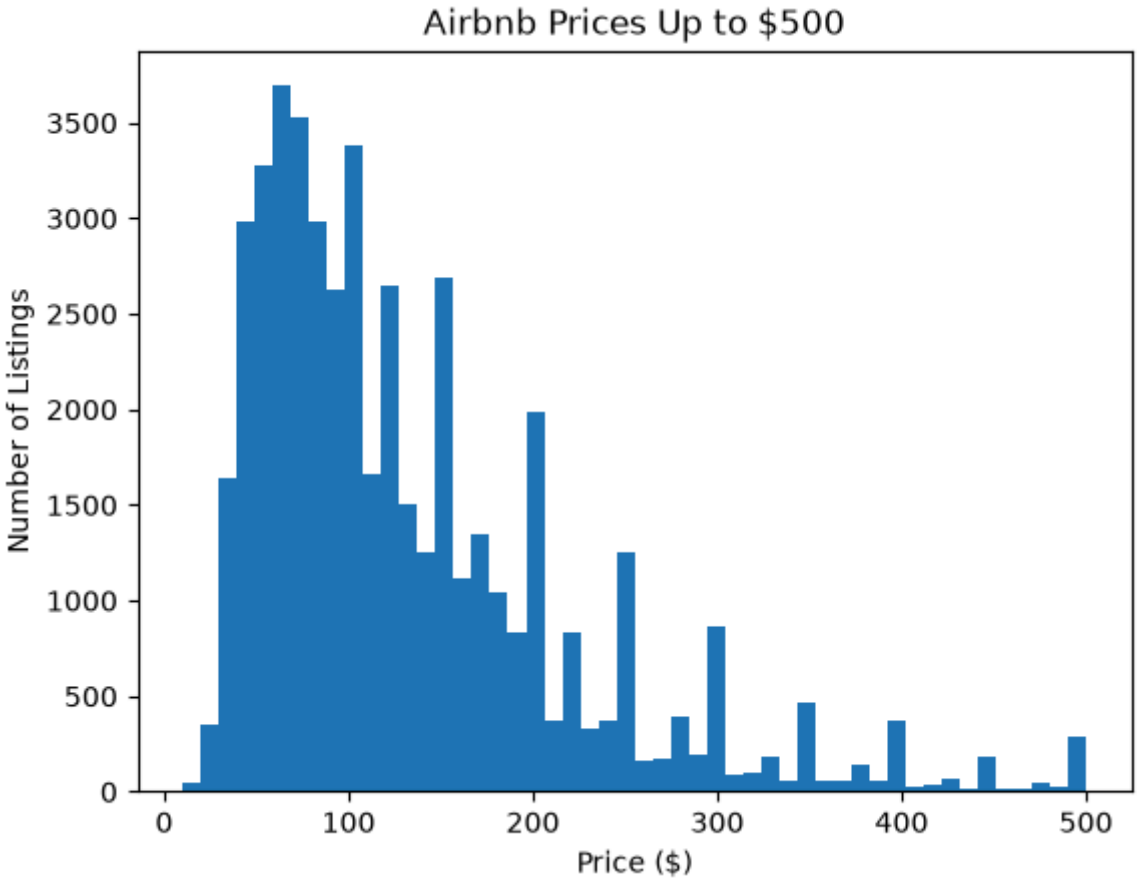
Eleven listings had a price of $0. They were excluded from price-based analysis by creating a separate cleaned dataset, leaving 48,884 listings for pricing analysis. The original dataset was preserved.

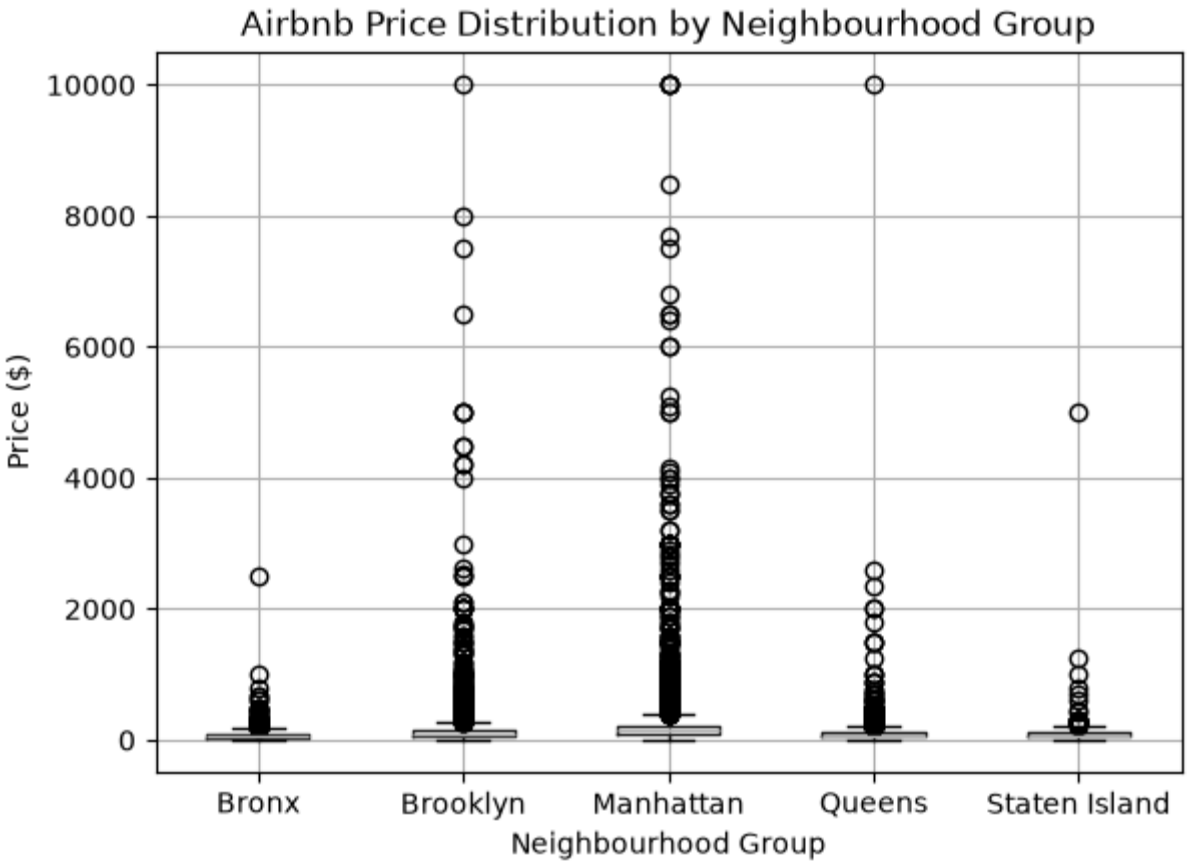
3. Exploratory Data Analysis — Python

The analysis examined price distributions, room types, neighbourhood groups, individual neighbourhoods, minimum-night requirements, reviews, and availability. Histograms, boxplots, scatter plots, and a correlation matrix were used to explore the data.

Key findings:

- Average listing price: $152.72; median listing price: $106.
- Prices were strongly right-skewed, with a small number of very expensive listings.
- Median prices by room type were $160 for entire homes/apartments, $70 for private rooms, and $45 for shared rooms.
- Manhattan had the highest average price among the five neighbourhood groups, at approximately $196.88.
- Manhattan had the highest median price within each room type.
- Neighbourhood comparisons were restricted to areas with at least 50 listings to reduce the influence of very small samples.
- Simple correlations between price and individual numerical variables were generally low.





4. SQL Analysis — MySQL

The dataset was imported into the airbnb_nyc database, in the ab_nyc_2019 table. The full dataset of 48,895 rows was loaded successfully. SQL queries were used to reproduce and extend the Python analysis.

Queries covered:

- Average prices and listing counts by room type.
- Price comparisons by neighbourhood group and room type.
- Neighbourhoods with the highest listing counts.
- Minimum nights, availability, and review statistics.
- Minimum, maximum, and average prices by neighbourhood group.

SQL techniques included SELECT, WHERE, COUNT, AVG, MIN, MAX, GROUP BY, ORDER BY, LIMIT, and multi-column grouping.

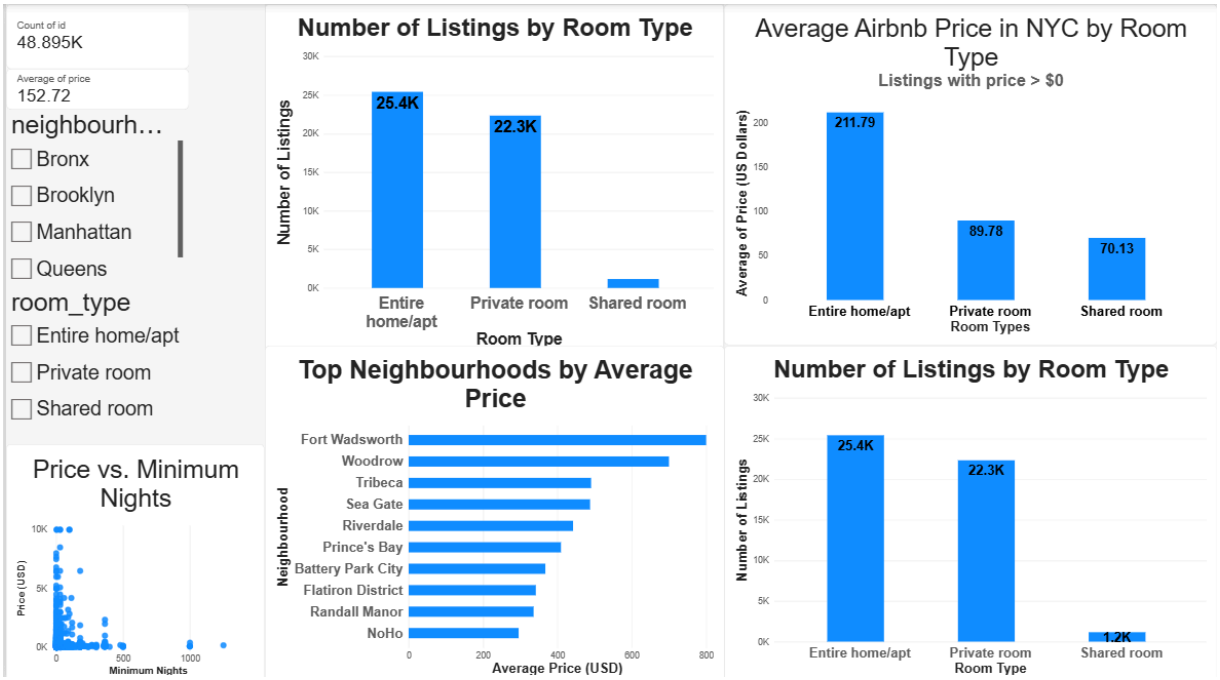
5. Power BI Dashboard

Created an interactive Power BI dashboard to present the main findings. It includes KPI cards for total listings and average price, with slicers for neighbourhood group and room type.

Dashboard visuals include:

- Average price by room type and neighbourhood group.
- Listing count by room type.
- Average minimum nights, availability, and reviews by room type.
- Price versus minimum nights.
- Top 10 neighbourhoods by average price.

Price-based visuals exclude the 11 listings with a price of $0. The dashboard allows users to explore pricing patterns by location and accommodation type.



6. Statistical Analysis and Regression

Multiple linear regression examined the association between price and room type, neighbourhood group, minimum nights, number of reviews, and availability. The initial model using raw prices had an R^2 of 0.092. Because prices were strongly right-skewed, a second model used log(price + 1), which achieved an R^2 of 0.472.

Main results from the log-price model:

- Compared with entire homes/apartments, private rooms were associated with approximately 54% lower prices and shared rooms with approximately 69% lower prices, holding other included variables constant.
- Compared with the Bronx, listings in Brooklyn were associated with approximately 29% higher prices, Manhattan with approximately 77% higher prices, and Queens with approximately 13% higher prices.
- The coefficient for Staten Island was not statistically significant.
- Minimum nights and number of reviews had small negative associations with log price, while availability had a small positive association.

These results describe statistical associations, not causal effects. The model does not explain all variation in prices, and its estimates should be interpreted with the limitations of the available variables in mind.

	price	minimum_nights	number_of_reviews	reviews_per_month	availability_365
price	1.000000	0.042805	-0.047926	-0.030575	0.081847
minimum_nights	0.042805	1.000000	-0.080080	-0.121644	0.144275
number_of_reviews	-0.047926	-0.080080	1.000000	0.549763	0.171975
reviews_per_month	-0.030575	-0.121644	0.549763	1.000000	0.185730
availability_365	0.081847	0.144275	0.171975	0.185730	1.000000

7. Limitations

- The dataset describes Airbnb listings in 2019 and does not represent the current market.
- Listing prices may differ from the final amount paid by guests.
- Extreme prices can affect averages.
- Statistics for neighbourhoods with few listings may be unreliable.
- The analysis identifies associations and does not establish causation.
- The regression model omits factors that may also influence prices.

8. Conclusion

Across the analyses, room type and location emerged as important factors associated with Airbnb listing prices in New York City. Entire homes/apartments were generally more expensive than private or shared rooms, and Manhattan showed the highest prices among the major neighbourhood groups. Python supported data cleaning and exploration, SQL enabled database-based analysis, Power BI provided an interactive presentation, and regression helped assess several variables simultaneously. Together, these stages form an end-to-end data analytics project.